Exercicis de dominis

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Calcula els següents dominis de funcions:

$f_1(x) = -3x^2 + 12$

Dom $f_1(x) = \mathbb{R}$

$f_2(x) = ln\,(3x + 1)$

$>0 \quad x > -\frac{1}{3}$ Dom $f_2(x) = (-\frac{1}{3}, +\infty)$

$f_3(x) = \frac{5}{x} = 5 \cdot x^{-1}$

Dom $f_3(x) = \mathbb{R} \setminus \{0\}$

$f_4(x) = \sqrt{3x + 2}$

$12 \geq 0 \quad x \geq \frac{-2}{3}$ Dom $f_4(x) = [-\frac{2}{3}, +\infty)$

$f_5(x) = \sqrt[3]{2x + 1}$

Dom $f_5(x) = \mathbb{R}$

$f_6(x) = \frac{(x-1)(x-2)}{x+1}$

Dom $f_6(x) = \mathbb{R} \setminus \{-1\}$

$f_7(x) = \frac{(x-1)(x-2)}{x-1}$

Dom $f_7(x) = \mathbb{R} \setminus \{1\}$

$f_8(x) = \frac{\sqrt{x+3}}{x+1}$

Dom $f_8(x) = [-3, -1) \cup (-1, +\infty)$

$13 \geq 0 \quad D \quad R$

$x \geq -3 \quad D = 0 \quad x = -1$

$f_9(x) = \frac{x+4}{\sqrt{2x+6}}$

Dom $f_9(x) = (-3, +\infty)$

$D = 0 \rightarrow x = -3$

-3

$f_{10}(x) = \frac{2}{x^2+1}$

Dom $f_{10}(x) = \mathbb{R}$

$f_{11}(x) = \frac{2}{x^2-1}$

Dom $f(x) = \mathbb{R} \setminus \{-1, 1\}$

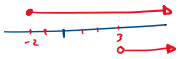
$D = 0 \rightarrow x = \pm 1$

$f_{12}(x) = \frac{\sqrt{x+2}}{\sqrt{x-3}}$

Dom $f_{12}(x) = (3, +\infty)$

$x + 2 \geq 0 \; ; \; x \geq -2$

$x - 3 \geq 0 \quad x \geq 3$



m) $f_{13}(x) = \frac{\sqrt{x^2-16}}{5x+1}$

n) $f_{14}(x) = \sqrt{\frac{2x+4}{x-3}}$

o) $f_{15}(x) = \sqrt{\frac{2x+4}{x^2-4}}$

p) $f_{16}(x) = \sqrt{-\frac{x-4}{3x-9}}$

q) $f_{17}(x) = ln\,(x^2 + 4)$ Dom $f_{17}(x) = \mathbb{R}$

$x^2 + 4 > 0 \quad x^2 + y = 0$

$x = -4$

r) $f_{18}(x) = \sqrt{x^2 - 4}$

s) $f_{19}(x) = \begin{cases} \frac{1}{x-4} & x > 5 \\ ln\,(-3x + 20) & x < 5 \end{cases}$

t) $f_{20}(x) = \begin{cases} \sqrt{x+1} & x > 0 \\ e^{-x} & x < 0 \end{cases}$

u) $f_{21}(x) = \begin{cases} \frac{x^2-1}{x^2-4} & x < -1 \\ \sqrt{3x+3} & -1 \leq x < 2 \\ \ln(3-x) & 2 \leq x \end{cases}$

w) $f_{23}(x) = 3^{\frac{1}{x} + \sqrt{2-x}}$

$2 - x \geq 0$

$2 \geq x$

$0 \notin$ Dom $f_{23}(x)$

Dom $f_{23}(x) = (-\infty, 0) \cup (0, 2]$

$= (-\infty, 2] \setminus \{0\}$

m) $f_{13}(x) = \frac{\sqrt{x^2-16}}{5x+1}$ Dom $f_{13}(x) = (-\infty, -4] \cup [4, +\infty)$

$x^2 - 16 \geq 0$

$x^2 - 16 = 0$

$x = \pm 4$

$D = 0 \quad 5x + 1 = 0$

$x = -\frac{1}{5}$

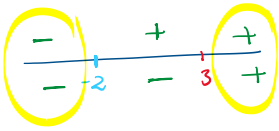
$f_{19}(x) = \begin{cases} \frac{1}{x-4} & x > 5 \\ ln(-3x+20) & x < 5 \end{cases}$ $(5, +\infty)$ $(-\infty, 5)$

$-3x + 20 > 0$

$20 > 3x$

$\frac{20}{3} > x$

Dom $f_{19}(x) = \mathbb{R} \setminus \{5\}$

n) $f_{14}(x) = \sqrt{\frac{2x+4}{x-3}}$ Dom $f_{14}(x) = (-\infty, -2] \cup (3, +\infty)$

$\frac{2x+4}{x-3} \geq 0$